

AI-Driven Intelligent Routing and Secure Autonomous Recovery in 6G-Integrated Disaster-Monitoring Sensor Networks

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Ph. D. Thesis in Computer Engineering

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- 1 Context and Motivation
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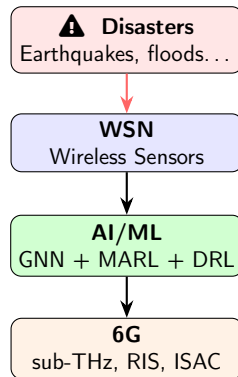
- 5 Article 2 — CALASH
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Context: Natural Disasters and Monitoring

- **7,348** major disasters (2000–2019)
4.2 B people affected, 1.23 M deaths, \$2,970 B in losses [UNDRR 2020]
- In 2023: 398 disasters, 95,000 deaths, ~\$380 B
- Climate change $\Rightarrow$ +10.5% intensity, +30% frequency of extreme precipitation [IPCC AR6]
- **Sendai Framework 2015–2030**: strengthen early warning systems

Need

Real-time monitoring via Wireless Sensor Networks (WSNs) — reliable, sustainable, and secure.



WSNs: Potential and Constraints

WSN Strengths

- Autonomous multi-hop data collection
- Coverage of large geographical areas
- Post-disaster operation capability
- Low per-unit cost

Severe Constraints

- Non-rechargeable batteries (~ 0.5 J)
- Limited RAM (16–256 KB)
- Restricted bandwidth
- Physical vulnerability in hazardous zones

6G Enabling Technologies (horizon 2030)

sub-THz	RIS	ISAC
Multi-Gbit/s	Passive wavefront ctrl	Sensing + comm.
O-RAN: near-RT RIC ≤ 10 ms $\Rightarrow$ AI in the control loop		

Three Fundamental Challenges



C1: Energy

- Routing = #1 factor for lifetime
- LEACH/QMIX/QTRAN: heuristics or $\sim 39K$ params/agent
- No MARL policy fusion exists



C2: Carbon

- Embodied $\gg$ operational ($10^6:1$)
- No protocol considers full lifecycle
- No modulation by grid intensity



C3: Security

- Rogue policy injection
- Out-of-distribution hallucinations
- Binary mechanisms only (allow/block)

Research Questions and Objectives

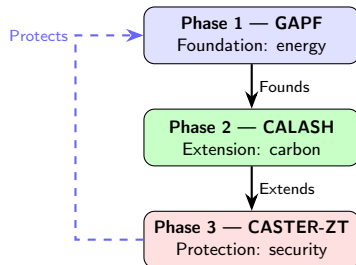
Main Research Question

How to design a complete intelligent routing protocol stack — energy-efficient, carbon-sustainable, and secure — for AI-native 6G-integrated disaster-monitoring sensor networks?

Specific Question		Specific Objective
Q1	PDR $\geq$ 98% via MARL + GNN fusion with $< 2,000$ params?	SO1 — GAPF: GCN+CAS fusion
Q2	Significant LCI reduction via 4 pillars + 6G PHY?	SO2 — CALASH: carbon-aware routing
Q3	Detection with 0 false positives + graduated decisions ≤ 10 ms?	SO3 — CASTER-ZT: zero-trust shield

Framework: Design Science Research (DSR)

- 1 Problem identification (UNDRR, IPCC)
- 2 Objective definition (Q1, Q2, Q3)
- 3 Artifact design (GAPF, CALASH, CASTER-ZT)
- 4 Demonstration (simulation)
- 5 Evaluation (>23,800 runs)
- 6 Communication (3 articles)



Common Experimental Protocol

Simulation Environment

- Python + PyTorch, PyMARL2
- Heinzelman first-order energy model
- Real-world data: Intel Lab, UK Carbon Intensity API, USGS ShakeMap
- Seismic model calibrated on Turkey–Syria 2023

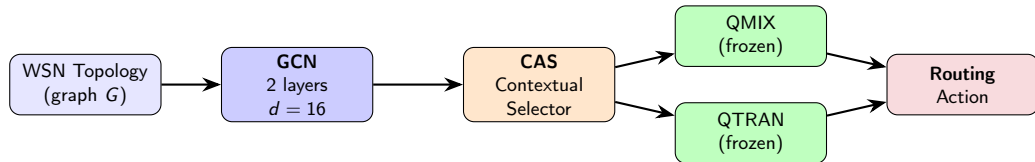
Statistical Rigor

- Article 1: 21,000 runs, 7 scenarios
- Article 2: 390 runs, 30 Monte Carlo seeds
- Article 3: 2,420 runs
- Total: > 23,800 runs
- Statistical tests: 95% CI, Wilcoxon, bootstrap

GAPF: Graph-Attentive Policy Fusion

Key Idea

Fuse complementary frozen MARL expert policies (QMIX & QTRAN) via a lightweight GNN encoder, instead of training a heavier monolithic network.



- GCN: encodes topology into a compact embedding
- CAS: contextual selection of the optimal expert
- Monotonic Q·K Attention Network for multi-objective reward scalarization
- Bi-level optimization: ES (outer loop) + MARL (inner)

GAPF: Why Fusion Works

QMIX / QTRAN Complementarity

- **QMIX**: $\partial Q_{\text{tot}} / \partial Q_i \geq 0 \Rightarrow$ tight coordination
- **QTRAN**: affine transforms $\Rightarrow$ non-monotonic interactions
- CAS selects expert by topological context

Monotonic Q·K Attention

- Multi-objective scalarization (energy, PDR, latency)
- Non-negative weights $\Rightarrow$ **Pareto monotonicity**
- Dominance $\Rightarrow$ higher scalar score
- Eliminates classical scalarization artifacts

“Expert Portfolio” Paradigm

Established in combinatorial optimization — applied 1st time to MARL routing in WSNs.

GAPF: Key Results

Metric	Baseline (QMIX)	GAPF	Gain
PDR (delivery ratio)	85–92%	$\geq$ 98%	+6–13 pts
Energy per packet	1.0×	0.29–0.50×	2–3.5× ↓
Trainable parameters	~39K/agent	1,473	26×
Training memory	3.2–13.8 GB	951 MB	3.4–14.5×
Model size	—	5.8 KB	Cortex-M4 OK



Contributions

- 1st MARL policy fusion framework via GNN for WSNs
- 26×
- Deployable on microcontroller without compression

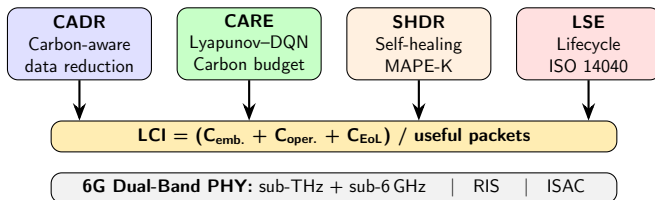
Validated on

- 7 scenarios (50–200 nodes)
- 21,000 runs
- Compared to LEACH, QMIX, QTRAN, ABC, PSO

CALASH: Carbon-Aware Lifecycle-Autonomous Self-Healing

Fundamental Insight

Embodied carbon dominates operational by $10^6:1$ in WSNs. Real lever: **maximize useful delivered packets** to amortize embodied carbon.



CALASH: The Four Pillars in Detail



CADR — Data Reduction

- Sensing ratio modulated by grid carbon intensity
- “Dirty” electricity: compress more
- “Clean” electricity: compress less



CARE — Carbon-Aware Routing

- Hybrid Lyapunov + DQN router
- Virtual carbon queue $\Rightarrow$ drift-plus-penalty
- **CDL Pareto bound**: provable convergence



SHDR — Self-Healing

- Autonomic MAPE-K loop
- Detects dead nodes post-disaster
- Rebuilds topology automatically



LSE — Lifecycle Engine

- ISO 14040 compliant
- Manufacturing + operation + end-of-life
- LCI: gCO_2eq / useful packet

CALASH: Key Results

Metric	Baseline (LEACH)	CALASH	Gain
Network lifetime	931 rounds	1,486 rounds	+60%
PDR	—	82.9%	—
LCI (gCO ₂ eq/pkt)	13.42	8.99	−33%
Physical layer	classical	6G dual-band sub-THz + RIS	

✓ Contributions

- 1st LCI metric for WSNs (ISO 14040)
- 1st 4-pillar protocol with Lyapunov guarantees
- Insight: reducing transmitted volume > reducing energy/packet

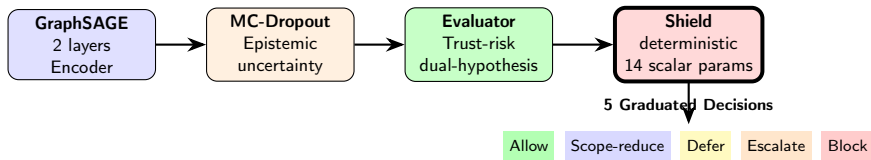
Validated on

- 390 runs, 30 Monte Carlo seeds
- Seismic model calibrated (Turkey 2023)
- UK Carbon Intensity API
- Compared to LEACH, HEED, EERP

CASTER-ZT: Zero-Trust Shielded Decision Control

Observation

AI in the control loop creates a **new attack surface**. Existing mechanisms (CPO, binary shielding) only offer *allow* or *block* — unsuitable for critical missions.



- NIST SP 800-207 compliant (zero-trust)
- Every actuation verified in real time
- 10 formal propositions proven
- Shield is **explainable** and **auditable**

CASTER-ZT: Formal Guarantees and Results

10 Formal Propositions

- Monotone conservatism
- Unauthorized-actuation exclusion
- Conformal coverage
- Graceful degradation
- Shield correctness & completeness

Deterministic Shield

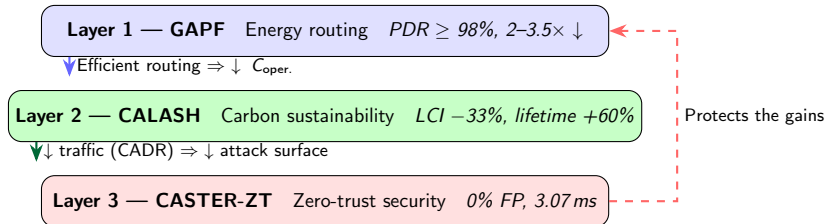
14 scalar params, non-learnable — last defense if neural parts compromised.

Metric	CASTER-ZT	vs. baseline
Rogue detection	74–98%	+14–28 pts
False positives	0%	Full elim.
ω_{rec}	0.733	3.7–5.2 $\times$ $\uparrow$
Latency/decision	3.07 ms	4.9 $\times$ $\downarrow$
O-RAN budget	≤ 10 ms	✓

Validated on

2,420 runs, 100 hex. cells. Compared to CPO, binary shield.

Protocol Stack Synergies



Mutual Reinforcement

The three layers reinforce each other: efficiency $\rightarrow$ sustainability $\rightarrow$ security. No trade-off.

Total Cost

$\sim 27K$ params ≈ 1 RNN agent, ~ 103 KB (FP32) $\Rightarrow$ Cortex-M7 OK. Full stack $<$ one QMIX agent budget.

Overall Quantitative Summary

Metric	Best baseline	Baseline val.	Our result	Gain
PDR	QMIX	85–92%	$\geq$ 98%	+6–13 pts
Energy / packet	QMIX	1.0×	0.29–0.50×	2–3.5× ↓
Parameters	RNN agent	~39K	1,473	26× ↓
Training memory	QMIX	3.2–13.8 GB	951 MB	3.4–14.5× ↓
Network lifetime	LEACH	931 rounds	1,486 rounds	+60%
LCI (CO ₂ /pkt)	LEACH	13.42	8.99	–33%
Rogue detection	Binary shield	60–70%	74–98%	+14–28 pts
False positives	CPO	2–8%	0%	Full elim.
Latency / decision	CPO	~15 ms	3.07 ms	4.9× ↓
Criteria covered (/12)	Best	4	12	3× ↑
Total test runs				> 23,800

TinyML Deployability

Framework	Params	Memory	Target	Compression
GAPF	1,473	5.8 KB	Cortex-M4	None needed
CALASH	~5K	19.5 KB	Cortex-M4	INT8 (→4.9 KB)
CASTER-ZT	~20K	78 KB	Cortex-M7	Distill.+INT8 (→20 KB)
Full stack	~27K	103 KB	Cortex-M7	Mixed quant.

Comparison

MCUNet deploys ~1M parameters on Cortex-M7.
Our stack: ~27K ⇒ **38× lighter**.

Implication

Deployable on microcontrollers costing a few dollars ⇒ accessible to developing countries (where 95% of deaths occur).

Six Original Scientific Contributions

- 1 **C1: First MARL policy fusion framework via GNN** for WSN routing
GNN+CAS paradigm transferable to any multi-agent system
- 2 **C2: Monotonic Q·K Attention Network** for multi-objective scalarization
Pareto monotonicity formally guaranteed — absent from existing methods
- 3 **C3: First Lifecycle Carbon Intensity (LCI)** metric for WSNs
ISO 14040 compliant — adoptable for any IoT protocol
- 4 **C4: First 4-pillar protocol with Lyapunov guarantees** for carbon budget
Provable convergence to near-optimal PDR AND bounded carbon budget
- 5 **C5: First zero-trust decision control** framework for AI-native networks
5 graduated decisions + 10 formal propositions + explainable shield
- 6 **C6: Demonstration of complementarity** between efficiency–sustainability–security
Integrated protocol stack: all 3 objectives reinforce each other



For Industry

- GAPF (5.8 KB): no costly gateway
- +60% lifetime $\Rightarrow$ $\downarrow$ maintenance
- LCI: sustainability reporting (CSRD/SEC)
- Auditable shield: IEC 62443



For Governments

- LCI: standardized IoT indicator
- Sendai indicators C-2, C-6
- Open access, no proprietary lock-in

SDG

Contribution

- | | |
|-----------|--|
| 9 | Resilient infrastructure (1,473 params) |
| 11 | Reduce losses ($\text{PDR} \geq 98\%$) |
| 12 | Sustainability (LCI ISO 14040) |
| 13 | Climate resilience (CADR) |
| 16 | Transparency (shield 0 FP) |



For Society

- $\text{PDR} \geq 98\%$: reliable alerts
- -33% LCI: toward carbon neutrality
- SHDR: post-disaster coverage

Acknowledged Limitations

Methodological

- ➊ **Simulation-based** — real data but no hardware
- ➋ **Simplified disaster model** — Gaussian, no cascades
- ➌ **Analytical 6G PHY** — no 3D ray tracing

Technical

- ➍ **Unvalidated integration** — not tested as stack
- ➎ **Scalability > 200 nodes** — not characterized
- ➏ **Adaptive adversaries** — static rogues only
- ➐ **No field validation** — no user study

Future Research Directions



Short (1–2 yr)

- Unified stack integration
- Hardware testbed (Zolertia, Coral, Jetson)
- Extended disaster models



Medium (2–4 yr)

- Federated learning
- O-RAN digital twins
- Multi-disaster / multi-site
- Formal cert. (Coq/Isabelle)



Long (4+ yr)

- Non-terrestrial (LEO, HAPS)
- LLMs as meta-controllers
- Standardization (ISO, 3GPP)

Transferability: Precision agriculture | Smart cities | Connected health | Drone swarms

Conclusion

Summary

This thesis proposed a **complete protocol stack** for intelligent routing in 6G-integrated disaster-monitoring WSNs, built around three complementary frameworks:



GAPF

MARL fusion via GNN
1,473 params, PDR $\geq$ 98%
Energy 2–3.5 $\times$ ↓



CALASH

4 pillars + 6G PHY
LCI –33%, lifetime +60%
CDL Pareto bound



CASTER-ZT

5 graduated decisions
0% FP, 3.07 ms
10 formal propositions

Key Message

AI, designed with **parsimony** (~27K params), **environmental awareness** (LCI), and **systematic verification** (zero-trust), is a reliable tool for critical environments.

Thank You!

Questions?

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Backup Slides

Convergence and Theoretical Guarantees

GAPF

- QMIX: converges within the class of monotonic functions (IGM)
- QTRAN: larger space via affine factorization
- CAS: *variance reducer* (Wolpert's stacking)
- Inference: $O(|E| \cdot d)$ for GCN

CALASH

- Lyapunov drift-plus-penalty optimization
- $O(1/V)$ convergence to optimum
- $V = 100$: -33% LCI + acceptable latency
- CDL bound: *simultaneous* convergence PDR + carbon budget

CASTER-ZT

- 10 formal propositions
- Proved by logical construction
- Monotonicity, exclusion, coverage
- Shield is correct and complete

Gap Analysis: Coverage of 12 Criteria

Criterion	LEACH	QMIX	CPO	Our stack
Energy-efficient routing	✓	✓		✓
Topology awareness (GNN)				✓
MARL policy fusion				✓
Pareto-monotonic scalarization				✓
Lifecycle carbon metric				✓
Carbon-aware data reduction				✓
Carbon budget guarantees				✓
Post-disaster self-healing				✓
6G physical layer				✓
Rogue policy detection			✓	✓
Graduated decisions (5 levels)				✓
Formal security guarantees				✓
Total (/12)	1	1	1	12

Theoretical Context

- Optimal Dec-POMDP resolution: **NEXP-complete**
- CTDE (QMIX, QTRAN): structured approximation reducing $|\mathcal{A}|^n \rightarrow \sum_i |\mathcal{A}_i|$
- GAPF: adds expert selection as an additional level
- $\text{PDR} \geq 98\%$ is not guaranteed *optimal* in the absolute sense, but constitutes a high-quality approximation

Positioning in the Value Decomposition Landscape

- QMIX $\rightarrow$ WQMIX $\rightarrow$ QPLEX $\rightarrow$ FOP: increasingly heavy architectures
- GAPF: **orthogonal** path — fusing lightweight experts via a topology-informed meta-controller
- “Portfolio” paradigm: well-established in combinatorial optimization, novel for MARL routing